

Extail: a Kinetic Inconspicuous Wearable Hair Extension Device

Yuiko Suyama

Tokyo Metropolitan University
6-6, Asahigaoka, Hino, Tokyo, Japan
suyama-yuiko@ed.tmu.ac.jp

Tetsuaki Baba

Tokyo Metropolitan University
6-6, Asahigaoka, Hino, Tokyo, Japan
baba@tmu.ac.jp

ABSTRACT

Wearable devices that present the wearer's information have been designed to stand out when worn, making it difficult to conceal their wear. Therefore, we have been working on developing and evaluating a dynamic expression of hair to realize the presentation of the wearer's information in an inconspicuous wearable device. In the precedents of hair interaction, the hairstyles and expressions to which the technique can be applied are limited. In this paper, we focus on the output mechanism and present Extail, a hair extension type device with a control mechanism that moves bundled hair like a tail. The results of a questionnaire survey on the correspondence between the movement of hair bundles and emotional expression were generally consistent with the results of the evaluation of tail devices in related studies.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Wearables, Body extension, Biomimetics, Communication design

ACM Reference Format:

Yuiko Suyama and Tetsuaki Baba. 2022. Extail: a Kinetic Inconspicuous Wearable Hair Extension Device. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3526114.3558648>

1 INTRODUCTION

Devices that present the wearer's state are often based on the motif of mammalian ears or tails[3][7][11]. These devices were designed on the premise that they are conspicuous because they give the wearer organs that are not inherently present in humans. However, there is no necessity for the medium of expression to have an appearance that makes the device stand out.

We work on developing and evaluating a dynamic expression of hair to realize the presentation of the wearer's information in an inconspicuous wearable device. In this paper, we present Extail, a hair extension type device with a control mechanism that moves bundled hair like a tail (Figure 1). Extail uses a stepper motor drive for quietness, and its operating noise is about 30-40dB.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3558648>



Figure 1: Wearing Extail

The use of hair as a medium of expression is based on the idea of adding a 'function' of dynamic expression to existing human parts, rather than adding borrowed 'forms' such as animal ears or tails. Furthermore, moving hair is often a means of emotional expression in cartoons and animations. This expression tends to a certain extent - for example, when a person is excited, the hair bundle flaps and swings, and when a person is sad, it hangs downward, and so on. In other words, there are already certain recognized links between moving hair and emotions, and it is reasonable to assume that we can use this shared understanding for communication.

2 RELATED WORK

Some research on head hair interaction deals with information input by touching the hair[6][9][10]. On the other hand, an example of a device that implements both input and output is an extension-type device with SMA braided into the braid [2]. However, in this system, the hairstyle is limited to braiding, and the movement is limited to extension and contraction. Other output expressions are limited to luminescence and robot arm-like movements. In the luminescence example, the optical fiber is made to look like hair, which is unnatural for hair material [1][5]. In addition, Young Suk Lee's wig is a work of art, leaving no room for the user to alter its shape. It is also large and conspicuous, making it unsuitable for daily use [4].

In other words, the dynamic output of hair is limited in its expression and in the hair shapes and materials to which the technique is applied. Hence, we focused primarily on the output mechanism, realizing the tail-like swing behavior of unbraided hairstyles such as ponytails. We emphasize that the device can be worn easily daily, just like enjoying fashion, and its appearance is so natural that one does not feel the device's presence.

3 IMPLEMENTATION

Figure 2 shows an overall view of the system. The main unit is specialized for output functions to make it compact and lightweight for wearing on the head. An external device performs sensing, and the main unit receives data from the external device via BLE communication. In this case, we used the heartbeat as an example of input information. We implemented a WatchOS app for heart rate sensing and an iOS app for data transfer to the main unit and manual input.

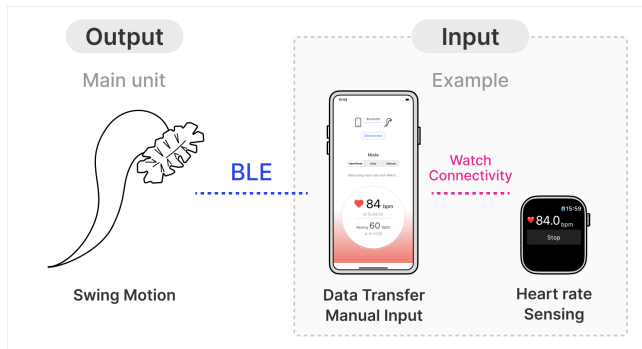


Figure 2: System Overview

3.1 Output Device

We prototyped tail mechanisms that mimic the tailbone of a mammal (Figure 3(a)). 2 or 3 strings are threaded through holes in joints, and the entire mechanism bends when this string is pulled. In order to make the device as small as possible, we tried to operate this string by winding it. However, we faced the challenge of string tangling within the winders. Therefore, we reduced the number of joints to the minimum number (two) that would allow the mechanism to function, shortened the entire mechanism, and reduced the length of the string by about 35 cm. Consequently, we employed this mechanism as a mechanism to move the root of the hair bundle rather than including it in the hair bundle.

Considering its quietness and operating speed, we use a stepper motor for the wind-up device. The wind-up device consists of a bobbin with a two-stage winding section attached to the shaft of a stepper motor (Figure 3(b)). When the motor turns, one side of the string is wound up, and the other is unwound. One of the two joints is fixed to the motor, and the hair is attached to the other with Velcro.

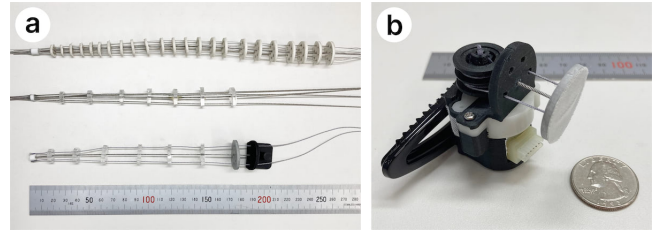


Figure 3: (a) Prototypes of the tail mechanism, (b) Adopted tail mechanism and winder

3.2 Input/Output Correspondence

We used the difference between resting heart rate (RHR) and heart rate (BPM) as the heart rate parameter and reflected it in the movement. We then referred to the following studies that evaluated the correspondence between tail device behavior and emotional expression in defining input-output correspondence.

Ashish Singh et al. state that a faster tail swing represents a state of high arousal, and a smaller tail swing represents a more positive state[8]. Haoran Xie et al. report that we perceive positive emotions such as relaxation and joy from large amplitudes and negative emotions from small amplitudes[11]. From these, at least, we can see that we have an image of intense movement for excitement and relaxed movement for rest. Therefore, we applied fast and small movements for higher heart rates and slow and large movements for lower heart rates. The impression on the amplitude magnitude will be verified through the impression evaluation in section 4.1 since the results of the two studies conflict.

3.3 Appearance

Figure 4 shows the appearance of the main unit of Extail. We used an alligator hair clip for the attachment part for easy attachment and removal. The control circuit and battery are hidden in the scrunchie, and this scrunchie is placed over the motor to look like a knot. The fabric of scrunchie and hair strands are removable, allowing the user to change their appearance according to their preference.



Figure 4: Appearance of Extail

Table 1: Description of operation during device noise measurement

Device	Interval	Operating Details
Extail	0-35 sec	The operating speed was fixed, and the amplitude gradually increased; The step interval was fixed at 2 ms. The rotational motion of the shaft was gradually changed from a continuous motion between 0-15 degrees to a continuous motion between 0-210 degrees.
	36-60 sec	The amplitude was fixed, and the operating speed was gradually increased; The rotational motion of the shaft was fixed with continuous motion between 0-210 degrees. The step interval was gradually varied from 2 ms to 10 ms.
necomimi	All	The device repeated action and standstill. The movements included in the measurement are slow, small movements of the ear and fast, large movements of the ear.

4 EVALUATION

4.1 Operating Noise

To evaluate the quietness of Extail, we measured the operating noise of Extail and necomimi[7]. We used the iOS app "Decibel X:dB Sound Level Meter"¹ for the measurements and placed the object 50 cm away from the iOS device. The ambient sound at the measurement site was around 28 dB. Since the loudness of both devices varied depending on how they moved, measurements were taken by operating the devices in a way that covered both large and small noise (Table 1). As a result, Extail's noise was about 30-40 dB, while necomimi's maximum noise was about 55 dB (Figure 5). Therefore, Extail reduced the maximum noise by about 15 dB compared to the servo motor-driven necomimi driven by servo motors.

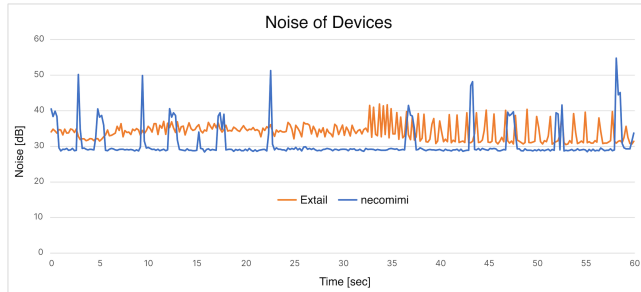


Figure 5: Results of noise measurements for Extail and necomimi

4.2 Impression

We conducted a questionnaire survey on nine people regarding the expression of waving hair bundles. Participants watched videos of eight movement patterns with various combinations of swing amplitude, speed, and turn waiting time and responded in an open-ended format to what the wearer appeared to be. See the our GitHub repository² for the videos used in the evaluation.

¹<https://apps.apple.com/us/app/decibel-x-db-sound-level-meter/id448155923>

²<https://github.com/mt-sumikko/extail-evaluation-202203/blob/main/README.md>

Participants agreed that small swings gave a negative impression, mainly fear, while fast and small swings gave a happy impression, such as "joy" and "happy."

Half of the participants indicated that irregular movements indicated a restless appearance. Two participants described the movements as mushy for tiny swings at a moderate speed.

For movements with large swings at moderate or higher speeds, some had a positive impression, and others had a negative impression. For slow swing, some had a calm impression, while others had an unsettled impression, and their impressions were conflicting.

They were unanimous in their positive impression of the fast, small swing movement. On the other hand, both positive and negative impressions were found for both small and large swings. Of these, the faster swing tended to lean more toward the positive. Including the variation in impressions of amplitude, these results were generally consistent with the evaluation results referenced in section 3.2.

We estimate that 3 to 4 patterns of movement are the limit for sharing a particular interpretation or distinguishing expressions on a single motor axis. However, most participants gave a different response for each movement, indicating that they may perceive subtle differences in the movement of the swaying hair bundles.

5 CONCLUSION AND FUTURE WORK

To present information with an inconspicuous device for wearing, we developed a hair extension type device, Extail, equipped with a control mechanism to swing bundled hair and evaluated the impression of its expression. This research demonstrated the possibility of utilizing hair in nonverbal communication.

Current devices present many challenges in terms of usability and diversity considerations. First, the device weighs about 130g, a little heavy to attach to one point on the head. In addition, the motor is too large to be covered by a small amount of hair. Furthermore, due to the size of the motor and clip, the hair bundle portion protrudes slightly too far from the wearing surface. Due to these issues, only back knot hairstyles are practical enough in appearance and usability. The back knot hairstyle has the fatal problem that it is challenging to see hair movement when viewed from the front of the wearer, making it meaningless to wear in some cases. In

order to reduce the burden on the head and expand the flexibility of styles, it is necessary to reduce the size and weight.

REFERENCES

- [1] Damien Brun and Jonna Häkkinä. 2021. *VäriWig: Interactive Coloring Wig Module*. Association for Computing Machinery, New York, NY, USA, 166–169. <https://doi.org/10.1145/3460421.3478832>
- [2] Christine Dierk, Sarah Sterman, Molly Jane Pearce Nicholas, and Eric Paulos. 2018. HäirIÖ: Human Hair as Interactive Material. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction* (Stockholm, Sweden) (*TEI '18*). Association for Computing Machinery, New York, NY, USA, 148–157. <https://doi.org/10.1145/3173225.3173232>
- [3] Shoko Kimura, Ayaka Fuji, and Kazunori Miyata. 2021. Impression evaluation of facial expression augmentation using a movable ear device linked to eyebrows (Original title in Japanese: Mayu to rendo suru kadou mimigata debaisu ni yoru hyojo kakucho no insho hyoka). In *Collection of Interaction 2021 Papers (Original title in Japanese: Interaction 2021 ronbunshu)*. Information Processing Society of Japan (IPSJ), Ishikawa, JPN, 505–510. <http://hdl.handle.net/10119/17057>
- [4] Young Suk Lee. 2018. Thou and I: Exploring Expressive Digital Interaction with Interactive Characteristic Wigs. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction* (Stockholm, Sweden) (*TEI '18*). Association for Computing Machinery, New York, NY, USA, 581–585. <https://doi.org/10.1145/3173225.3173311>
- [5] Bing Li, Dawei Zheng, Yujia Lu, Fangtian Ying, and Cheng Yao. 2017. LightingHair Slice: Situated Personal Wearable Fashion Interaction System. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) (*CHI EA '17*). Association for Computing Machinery, New York, NY, USA, 1824–1828. <https://doi.org/10.1145/3027063.3053093>
- [6] Marie Muehlhaus, Jürgen Steimle, and Marion Koelle. 2022. Feather Hair: Interacting with Sensorized Hair in Public Settings. In *Designing Interactive Systems Conference* (Virtual Event, Australia) (*DIS '22*). Association for Computing Machinery, New York, NY, USA, 1228–1242. <https://doi.org/10.1145/3532106.3533527>
- [7] neurowear. 2012, 2021. *necomimi*. Retrieved July 16, 2022 from <https://neurowear.com/necomimi/>
- [8] Ashish Singh and James E. Young. 2013. A Dog Tail for Utility Robots: Exploring Affective Properties of Tail Movement. In *Human-Computer Interaction – INTERACT 2013*, Paula Kotzé, Gary Marsden, Gitte Lindgaard, Janet Wesson, and Marco Winckler (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 403–419. https://doi.org/10.1007/978-3-642-40480-1_27
- [9] Hiroaki Tobita and Takuya Kuzi. 2012. SmartWig: Wig-Based Wearable Computing Device for Communication and Entertainment. In *Proceedings of the International Working Conference on Advanced Visual Interfaces* (Capri Island, Italy) (*AVI '12*). Association for Computing Machinery, New York, NY, USA, 299–302. <https://doi.org/10.1145/2254556.2254613>
- [10] Katia Vega, Marcio Cunha, and Hugo Fuks. 2015. Hairware: The Conscious Use of Unconscious Auto-Contact Behaviors. In *Proceedings of the 20th International Conference on Intelligent User Interfaces* (Atlanta, Georgia, USA) (*IUI '15*). Association for Computing Machinery, New York, NY, USA, 78–86. <https://doi.org/10.1145/2678025.2701404>
- [11] Haoran Xie, Kento Mitsuhashi, and Takuma Torii. 2019. Augmenting Human With a Tail. In *Proceedings of the 10th Augmented Human International Conference 2019* (Reims, France) (*AH2019*). Association for Computing Machinery, New York, NY, USA, Article 35, 7 pages. <https://doi.org/10.1145/3311823.3311847>